

Learned representations

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can trace an input through a hidden representation and a decision rule.

I can identify distinctions that a representation preserves or loses.

I can separate a useful constructed feature from evidence of successful learning.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Trace an input through a hidden representation and a decision rule

Identify distinctions that a representation preserves or loses

Separate a useful constructed feature from evidence of successful learning

Retrieve what you already know

Why should a debugging trace inspect scores before ReLU as well as final activations?

My starting explanation

This week's end goal

By the end, I can explain learned representations using a traced example and a stated limitation.

2. Notice the evidence

Consider fictional rectangles with input [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise. A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

Define two hidden units: $h1 = \text{ReLU}(\text{width} - \text{height})$ and $h2 = \text{ReLU}(\text{height} - \text{width})$. Their weight rows are [1, -1] and [-1, 1], with zero biases. The representation $h = [h1, h2]$ records excess width or excess height. Predict 1 when $h1 > 0$; otherwise predict 0.

Input · Hidden representation · Prediction

Input: [4, 2]

Hidden representation: [2, 0]

Prediction: 1

Input: [2, 4]

Hidden representation: [0, 2]

Prediction: 0

Input: [3, 3]

Hidden representation: [0, 0]

Prediction: 0

Input: [5, 3]

Hidden representation: [2, 0]

Prediction: 1

These weights were chosen by the lesson author. They demonstrate a representation that could be

implemented by a network, but no training run has established that a model learned it. Learning would require adjusting parameters using training data and recording the procedure.

List the given facts and one fact that is not supplied.

3. See how it works

A prediction head is the computation that reads the hidden vector and produces an output. Keeping it separate lets us inspect what information is available before making a decision. This example assumes two finite, positive measurements in cm; it is not a general input validator.

```
def represent(width, height):  
    return [max(0, width - height),  
            max(0, height - width)]  
  
def predict(hidden):  
    return int(hidden[0] > 0)  
  
examples = [(4, 2), (2, 4), (3, 3), (5, 3)]  
for width, height in examples:  
    hidden = represent(width, height)  
    print([width, height], hidden, predict(hidden))  
  
assert represent(4, 2) == represent(5, 3)  
assert predict(represent(3, 3)) == 0
```

The raw difference width - height can be recovered as $h_1 - h_2$. The original width and height cannot in general be recovered: many rectangles share a difference. Two hidden components do not guarantee preservation of two input components.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact predictions correct.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: The head keeps both ordered hidden values instead of their sum.

Prediction: It can distinguish [3, 0] from [0, 3] in this example.

Reason: The representation retains the direction of brightness difference when coordinates stay separate.

5. Words that unlock the idea

Word · Meaning

Word: Representation

Meaning: An encoding of an example used by another computation

Word: Hidden activation

Meaning: A value produced inside a network after applying an activation function

Word: Collision

Meaning: Two different inputs receiving the same representation

Word: Learned feature

Meaning: A feature whose computation depends on parameters adjusted using training data

Use two terms to explain a step in this week's example.

6. Follow a worked example

Different rectangles can share a hidden vector

Input in cm · Hidden vector · Area in square cm

Input in cm: Width 4 height 2

Hidden vector: [2, 0]

Area in square cm: 8

Input in cm: Width 5 height 3

Hidden vector: [2, 0]

Area in square cm: 15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learned representations

See when a representation preserves a distinction and when it destroys it.

The facts we will use

Toy input [left, right]; representation $h=[\text{ReLU}(\text{left}-\text{right}), \text{ReLU}(\text{right}-\text{left})]$. $A=[5, 2]$; $B=[2, 5]$; target is left-brighter versus right-brighter.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes $[3, 0]$, with left brighter.



2. Encode B

B becomes $[0, 3]$, with right brighter.



3. Compare collapsed summary

If the head receives only $h_1 + h_2$, both become 3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Encode A

What happens and why: A becomes $[3, 0]$, with left brighter.

Step: 2. Encode B

What happens and why: B becomes $[0, 3]$, with right brighter.

Step: 3. Compare collapsed summary

What happens and why: If the head receives only $h_1 + h_2$, both become 3 and the target distinction is lost.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A hidden representation decides which distinctions remain available to an output head. These two hidden coordinates preserve whether the left or right side is brighter. Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical inputs; retain the distinction or change the task.

What this example does not establish

This is a hand-chosen representation, not evidence that a trained image model actually learned these units.

Read the labelled picture: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: The head keeps both ordered hidden values instead of their sum. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

A representation can be sufficient for one task and insufficient for another. Our hidden vector supports the wider-than-height rule for every valid input under the stated formula. It fails to identify exact area for all rectangles because a collision can contain different areas. No more complicated deterministic head can recover information that is absent from its input.

Compare a second encoding $s = \text{width} + \text{height}$. Rectangles $[4, 2]$ and $[2, 4]$ both give $s = 6$ but have different wider-than-height targets. A head receiving only s must fail on at least one of them. Useful features depend on the question, not just the number of encoded values.

One repair for area is to retain the original measurements. Another is to append s to h . Since $d = h_1 - h_2$, $\text{width} = (s + d) / 2$ and $\text{height} = (s - d) / 2$. This works with exact valid measurements; it does not mean every added feature improves generalisation on noisy data.

Locate the earliest unsupported step

The alternative representation is $s = \text{width} + \text{height}$. Use $[4, 2]$ and $[2, 4]$ to decide whether s alone always supports the wider-than-height task.

State one repair and the checks you would rerun.

8. Practise together

Encode $[8, 3]$, $[3, 8]$ and $[4, 4]$ with the lesson layer. Apply its prediction rule and explain the square boundary.

Encode $[6, 4]$ and $[7, 5]$. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Explain your prediction to a partner before comparing results.

Practise with less help: Prove an information loss

Use workbook W2 for the connected example “Learned representations”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Learned representations. Proposed explanation: “Two hidden values preserve both input measurements”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact predictions correct.

```
def represent(width, height):  
    return [max(0, width - height),  
            max(0, height - width)]  
  
def predict(hidden):  
    return int(hidden[0] > 0)  
  
examples = [(4, 2), (2, 4), (3, 3), (5, 3)]  
  
for width, height in examples:  
    hidden = represent(width, height)  
    print([width, height], hidden, predict(hidden))
```

```
assert represent(4, 2) == represent(5, 3)

assert predict(represent(3, 3)) == 0
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its prediction rule and explain the square boundary.

The alternative representation is $s = \text{width} + \text{height}$. Use [4, 2] and [2, 4] to decide whether s alone always

supports the wider-than-height task.

For $h = [2, 0]$ and $s = 8$, recover width, height and area. Does this algebra prove that a training program learned the representation or that it will work well on noisy measurements?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled learned representations evidence sheet to Image-pattern Network Investigation. Show how you can trace an input through a hidden representation and a decision rule. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

In a trained network, hidden features depend on fitted weights and may not have simple human names. A strong output result does not by itself show that a unit represents a concept such as shape or fairness. Inspect activations on deliberately varied examples, compare with a baseline, and limit conclusions to the evidence.

A teammate calls the hand-chosen rectangle layer a learned area detector because all four wider-than-height predictions are correct. Identify two problems with this claim and propose one decisive area check.

When testing a learned representation, fit preprocessing and parameters on training data, use validation data for choices, and reserve the final test for the declared final evaluation. Compare representations with matched data, head rules and training budgets, and report collisions and failures. Next week makes the axis and channel meanings of image tensors explicit.

For $h = [2, 0]$ and $s = 8$, recover width, height and area. Does this algebra prove that a training program learned the representation or that it will work well on noisy measurements?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Different rectangles can share a hidden vector

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Hidden vector: [2, 0]

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Input in cm: Width 5 height 3

Hidden vector: [2, 0]

Area in square cm: 15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Audit a new representation independently

A different layer uses $g = [\text{ReLU}(\text{width} - \text{height} - 1), \text{ReLU}(\text{height} - \text{width} - 1)]$. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height. Can any deterministic head receiving only g classify both correctly?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A different layer uses $g = [\text{ReLU}(\text{width} - \text{height} - 1), \text{ReLU}(\text{height} - \text{width} - 1)]$. Encode $[4, 3]$ and $[3, 4]$. State their targets for width strictly greater than height. Can any deterministic head receiving only g classify both correctly?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: See when a representation preserves a distinction and when it destroys it.

By the end, I can · How I will show it

By the end, I can: Trace an input through a hidden representation and a decision rule

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify distinctions that a representation preserves or loses

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Separate a useful constructed feature from evidence of successful learning

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Trace an input through a hidden representation and a decision rule

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.